Arista Access Node Failure Resolution SOP: Jayanagar Metro Site - Node ID: JAY-ACC-55

Node: JAY-ACC-55 | **Vendor:** Arista | **Severity:** Warning | **Alarm Code:** ALARM_5427

Description

This Standard Operating Procedure outlines the diagnostic and resolution procedures for a Warning Failure on the Arista Access Node JAY-ACC-55, located at the Jayanagar Metro Site. The failure is caused by Hardware/Environmental Faults, specifically ASIC memory parity errors, line card crashes, or power supply redundancy loss. This SOP adheres to the command-line interface (CLI) dialect of Arista equipment. Access nodes are critical components in the telecommunication network, responsible for providing Layer 2 and Layer 3 connectivity to network users. In the Arista EOS (Extensive Operating System), the access node role is enabled by the "switching" feature. Failures in the access node can lead to network congestion, packet loss, and other degradation in network performance. In the given scenario, the failure is identified by the ALARM_5427 code, indicating a Warning condition. This SOP aims to guide the network administrator through the diagnostic and resolution process, ensuring minimal network downtime and optimal network performance.

Pre-requisite Checks

- Ensure remote hands are dispatched to the Jayanagar Metro Site to physically access the equipment.
- Verify the redundant link status to identify any potential issues.
- Check the power supply redundancy to confirm if it is functioning correctly.

Potential Root Causes

Cause	Probability
ASIC memory parity errors	High
Line card crashes	Medium
Power supply redundancy loss	Low

Diagnostic Commands & Expected Outputs

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Command: show system internal switch database, Expected Output: Switch database information, including the current state of the switch fabric and any errors encountered., Description: This command displays the switch database information, allowing the network administrator to identify any potential issues with the switch fabric.

Command: show switch fabric detail, Expected Output: Detailed information about the switch fabric, including the current state of the fabric and any errors encountered., Description: This command displays detailed information about the switch fabric, enabling the network administrator to diagnose any issues with the fabric.
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Command: show environment temperature, Expected Output: Current temperature reading of the equipment, including any alarm conditions., Description: This command displays the current temperature reading of the equipment, allowing the network administrator to identify any potential environmental issues.
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Resolution Steps

Step 1: Step: Clear any existing alarms, Command: clear alarm, Description: Clear any existing alarms to prevent them from interfering with the diagnostic and resolution process.

Step 2: Step: Replace the faulty line card, Command: replace line-card, Description: Replace the faulty line card with a new one to restore normal operation of the access node.

Step 3: Step: Verify power supply redundancy, Command: show power supply detail, Description: Verify the power supply redundancy to confirm if it is functioning correctly and providing adequate power to the equipment.

Post-Check Verification

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Command: show system internal switch database, Expected Output: Switch database information, indicating normal operation., Description: Verify that the switch database information displays normal operation, confirming that the resolution has been successful.

Command: show switch fabric detail, Expected Output: Detailed information about the switch fabric, indicating normal operation., Description: Verify that the detailed information about the switch fabric displays normal operation, confirming that the resolution has been successful.
```

Roll-back Procedures

- Step: Restore original configuration, Command: copy running-config startup-config, Description: Restore the original configuration to ensure that the access node returns to its previous state.

- Step: Reboot the access node, Command: reload, Description: Reboot the access node to ensure that all changes are rolled back and the access node returns to its previous state.